Graph Sketching: Hyperbola

Questions 1–7

Question 1. State the asymptotes of each of the following curves:

a) $\frac{x^2}{16} - \frac{y^2}{4} = 1$

b) $x^2 - 9y^2 = 36$

c) $x^2 - 2y^2 = 16$

Question 2. Sketch each of the following curves, label each intersection with the axes clearly and label any asymptotes in your sketch.

a) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

b) $\frac{x^2}{2} - \frac{y^2}{5} = 1$

c) $x^2 - 5y^2 = 5$

d) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a > b$

e) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a < b$

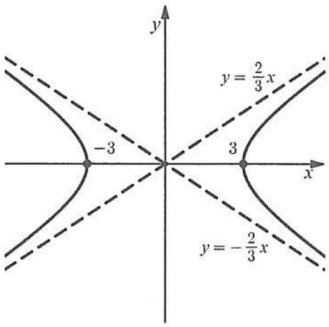
f) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a = b$

Question 3. For constants a and b, each of the curves below are of the form

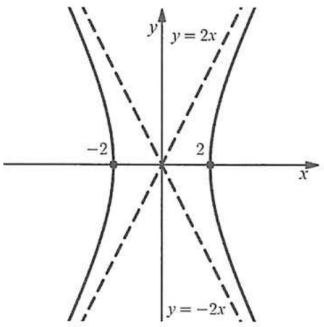
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

In each case find the value of a and b

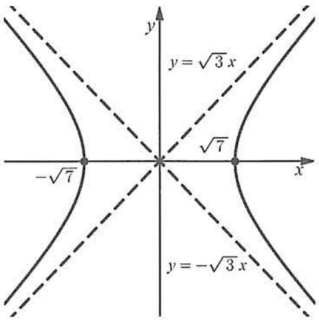
a)



b)



c)



Question 4. For a constant a, a hyperbola is given by

$$\frac{x^2}{a^2} - \frac{y^2}{2a^2} = 1$$

a) Write down the equations of the asymptotes of the curve.
b) Hence, sketch the curve,

Question 5. For constants a and b, a hyperbola is given by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

It passes through $P(-3, 0)$ and $Q(6, \sqrt{6})$

a) Find the value of a and b

b) Hence, sketch the curve.

Question 6. For constants a and b, a hyperbola is given by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

It passes through $P(2, -\sqrt{3})$ and $Q(-3, 3\sqrt{2})$

a) Find the value of a and b

b) Hence, sketch the curve.

Question 7. Find the equation of the curve below in the form $x^2 - py^2 = q$ where p and q are constants to be found.

